UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

0504609044

PHYSICS **5054/04**

Paper 4 Alternative to Practical **May/June 2008**

1 hour

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper-clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **9** printed pages and **3** blank pages.

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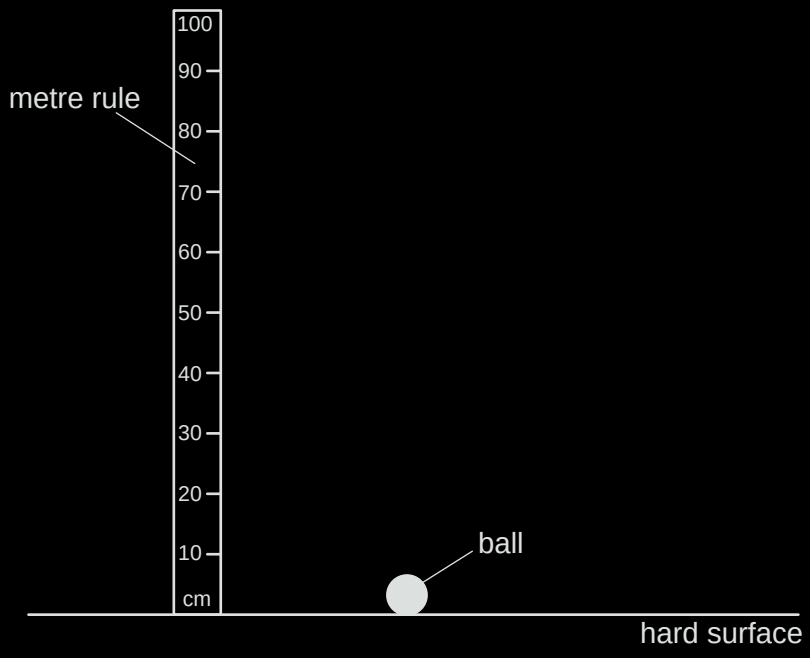


Fig. 1.1

(a) The ball is dropped from a height of 1.00 m above the hard surface.

State which part of the ball should be used when measuring its height above the surface.

..... [1]

(b) After the first bounce, the ball reaches a maximum height of 0.66 m.

On Fig. 1.1,

(i) draw the ball at a height of 0.66 m,

(ii) mark where you would position your eye to measure this height. [2]

(c) Explain

(i) why the maximum height *h* of the ball after the first bounce is difficult to measure,

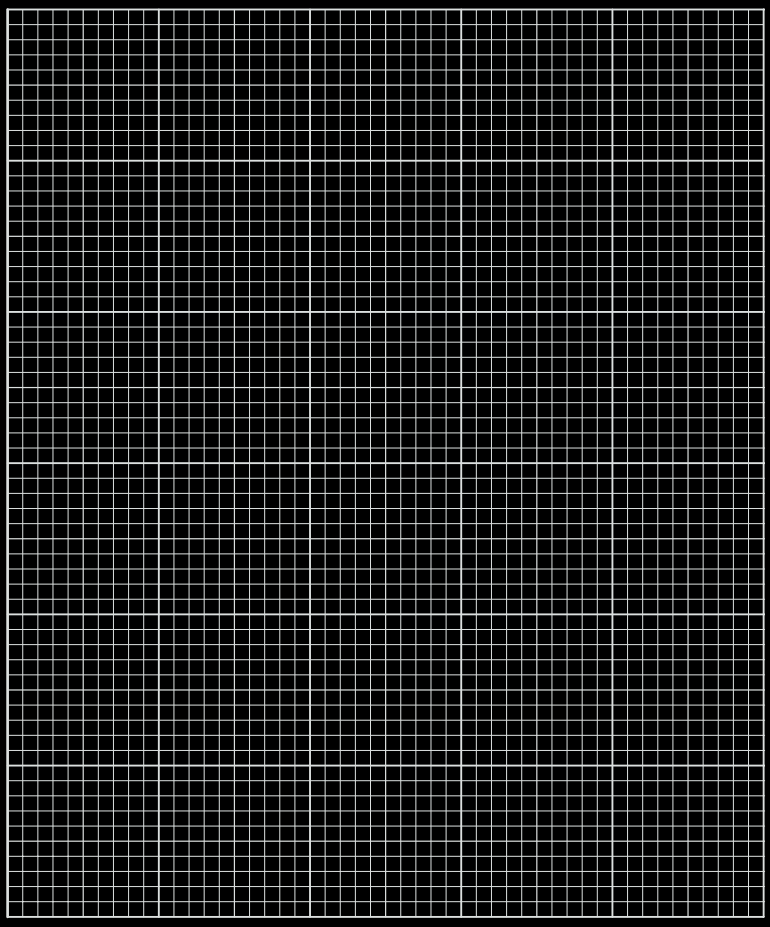
.....

..... [1]

(ii) how this height can be measured more accurately by two students working together.

.....

..... [1]



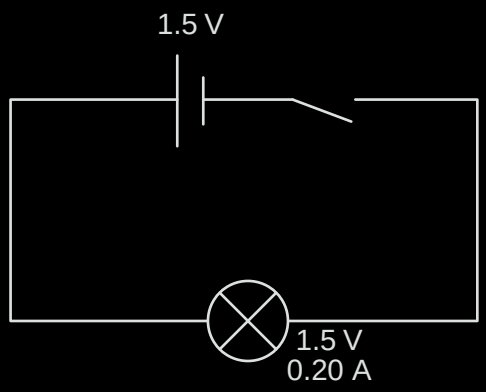


Fig. 2.1

(a) When the switch is closed the lamp does not light up.

Explain, with the aid of a diagram, how to use a voltmeter to find out whether the cell has run down.

.....

..... [2]

(b) Suggest three other possible faults in the circuit that might prevent the lamp from lighting.

1. .....

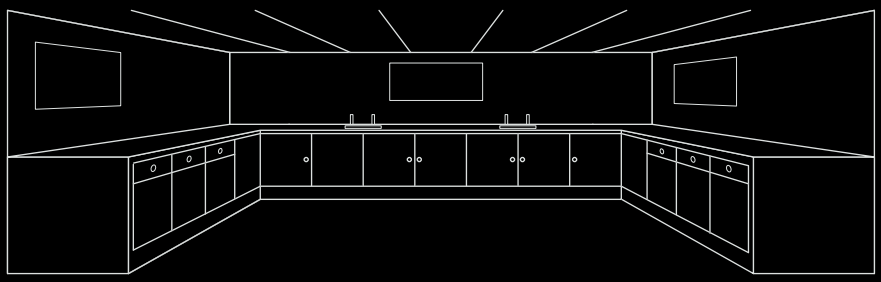
..... [1]

2. .....

..... [1]

3. .....

..... [1]



(a) State

(i) the measuring instrument used .....

..... [1]

(ii) the measurements taken .....

..... [1]

(iii) how the volume of the air is calculated .....

..... [1]

(b) State two possible sources of error in their answer.

1. .....

.....

2. .....

..... [2]

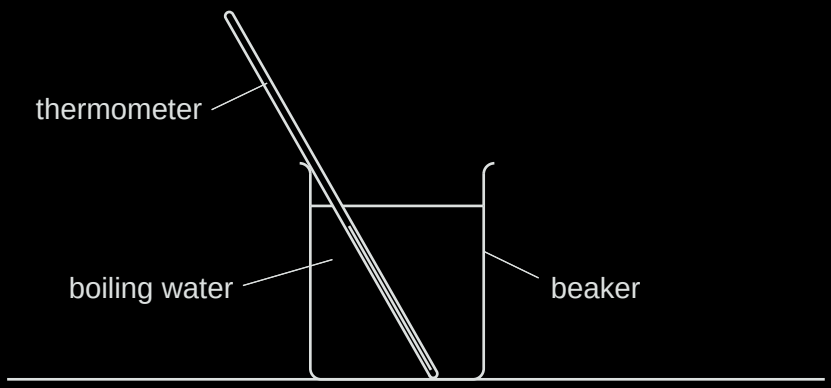


Fig. 4.2

(i) State and explain which thermometer from Fig. 4.1 is the most suitable for this experiment.

.....

.....

.....

..... [3]

(ii) Describe how the thermometer is used in this experiment to obtain accurate readings.

.....

.....

..... [2]

(c) Thermometer D in Fig. 4.1 is used to measure the temperature of a person.
Fig. 4.3 shows a modern forehead thermometer. It is a thin flexible plastic strip that is placed on the forehead. The colour of the numbers changes to show the temperature.

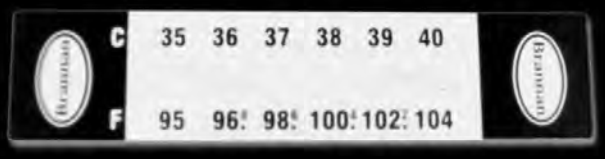


Fig. 4.3

State one advantage of this thermometer when taking the temperature of a young child.

.....

..... [1]

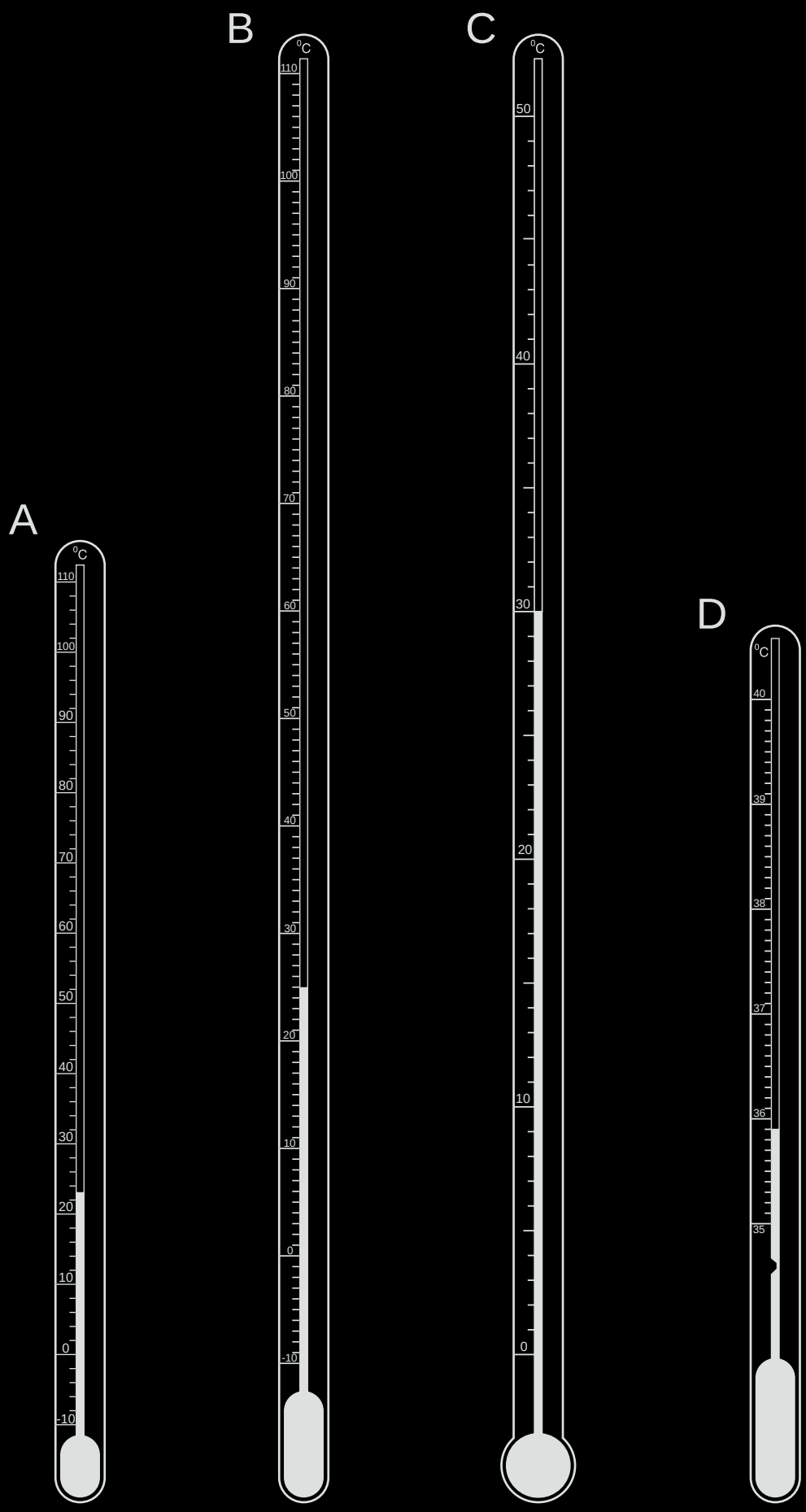

A
°C
110
100
90
80
70
60
50
40
30
20
10
0
-10
B
°C
110
100
90
80
70
60
50
40
30
20
10
0
-10
C
°C
50
40
30
20
10
0
D
°C
40
39
38
37
36
35

Fig. 4.1